Enabling a Power Electronics Grid

Prof Deepak Divan, Director – Center for Distributed Energy, Georgia Tech
Invited Keynote – NSF Workshop

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GT Center for Distributed Energy

Creating holistic solutions in electrical energy that can be rapidly adopted and scaled

Platform Initiatives

Grid Asset Augmentation

13 kV/50 kVA FUT
13 kV 1 MW Power Router
67 MVA Modular LPT
Improving Grid Resiliency
Smart Wires
Meshed Grid VVC

Energy Access in Emerging Markets

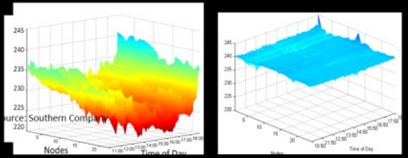
'Exponential' Tech
Self Organizing Nano Grid
Pay-Go Smart Meter
Low Cost DA for Grids
Ad-Hoc Bottom-Up Grids
Empower a Billion Lives

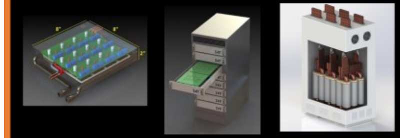
Next Generation Grid Power Electronics

5 kV DC Grid Building Block
7.2 kV 50 kVA Grid Connected SST
4 kV MVSI for Large PV Farms
Triports for PV/Storage/Grid
MVSI with Integrated Storage
Microgrid-Grid Interface Device

Decentralized Grid Control Techniques & Markets

Grid Edge Volt VAR Control
Collaborative Control
High PV Integration
DER Micro grid Impact
Self-Pricing Island Grids
Virtual Power Plants



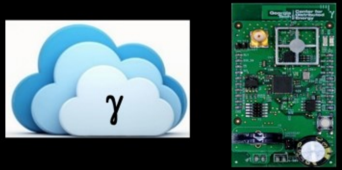


Next Generation Industrial Power Electronics

Industrial CVR Energy Efficiency
100 kVA EV Drive System
25-500 kVA Isolated Drives
Energy Hub – DC Fast Charging
Programmable Load/Source
Data Center Power Sources

Global Asset Monitoring Management & Analytics (GAMMA)

Low-Cost Communications
Cyber-Security
Data Management
AMI Data Analytics
Global Sensor Networks
Cloud Based GAMMA System



Future Grid – Drivers of Change

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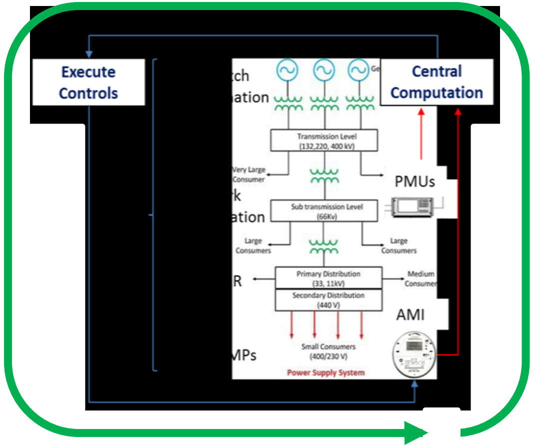
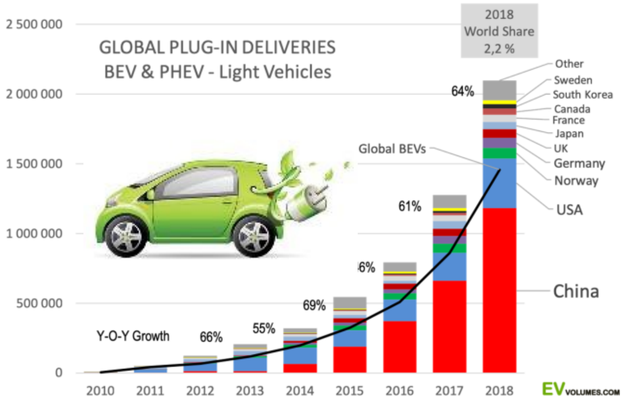
Today: Centralized, Passive & Rigid → **Tomorrow: Decentralized, Dynamic & Resilient**

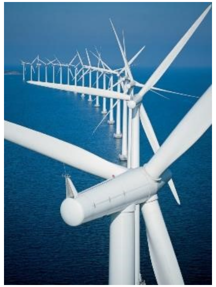


- Today's bulk power system is centrally controlled, managed by an ISO, with concepts such as LMP & market functions allowing complex operation
- Grid is rapidly changing with Exponential Technologies – distributed, decentralized, non-dispatchable, autonomous, fractal, economical – traditional control paradigms breaking down
- High DER penetration is raising concerns about loss of inertia, degraded stability & decentralized control – what does a PE dominant grid look like, how does it behave?



Fast Moving Exponential Technologies

Computation, PV solar, wind, EV, power semis, storage, microcontrollers, sensors, IoT, communication technologies, online services, social media, mobile pay, block-chain, cloud, autonomous control, deep learning



PV & WIND

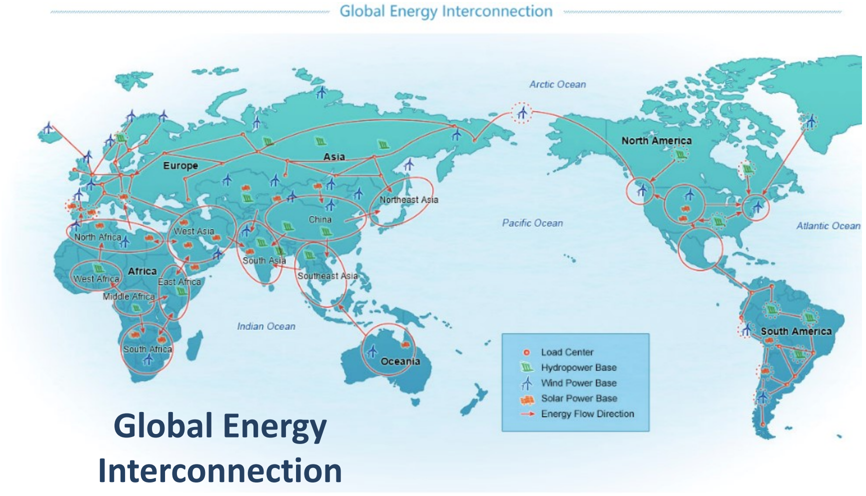
2019: Wind + 4 hours storage: $24/MWHr
PV + 4 hours storage: $32/MWHr

PROSUMERS

Power Electronics on the Grid*…traditional view*

- The AC grid is passive, with control typically realized using control of generator power/voltage & network topology
- The first major use of power electronics on the grid was in HVDC systems to transfer power over long distances
- The second application was with Flexible AC Transmission Systems (SVC and STATCOM) for dynamic volt-VAR control
- As wind and PV solar energy emerged, they were treated as ‘grid-following’ loads, assuming little impact on the grid
- As DERs grew, interaction with the grid increased, almost causing grid collapse in some cases – led to LVRT standards
- Growing interest in the possibility of a global HVDC link for transcontinental interconnections to enable clean energy

HVDC/HVDC Light



Global Energy Interconnection

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PE Control on the Grid – New Solutions

Distributed Grid Control … *Real Examples*

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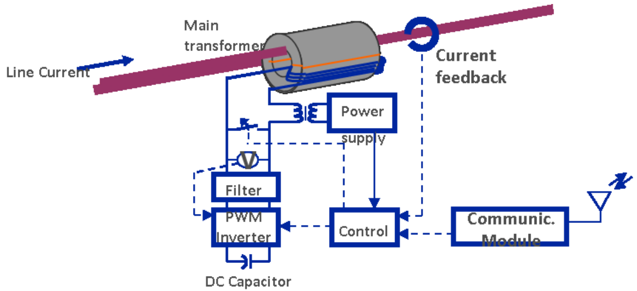
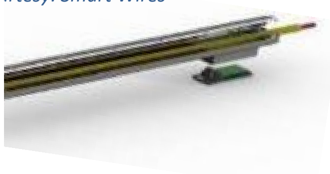
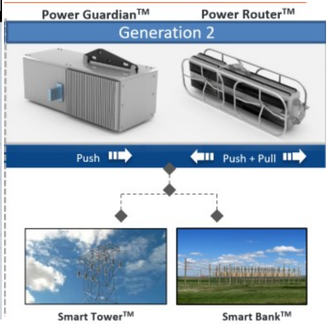
REIMAGINE THE GRID

Transmission Power Flow Control … *Smart Wires*

- Real-time distributed control of transmission power flows
- Smart Wires showed a new way to control power flows on the grid
- Smart Wires is seeing good traction with deployments worldwide
- Proven on 230-350 kV/2000 A systems with 65 kA fault current



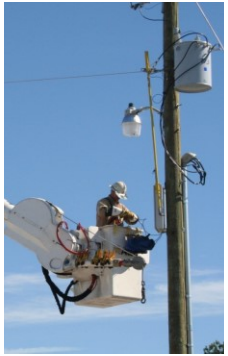




Decentralized Grid-Edge Volt-VAR Control … *Varentec*

- Real-time decentralized grid-edge VVC
- Stabilizes voltage profile across entire feeder
- Approved by Hawaii and Colorado PUC
- Increases PV penetration by >100% (HECO)

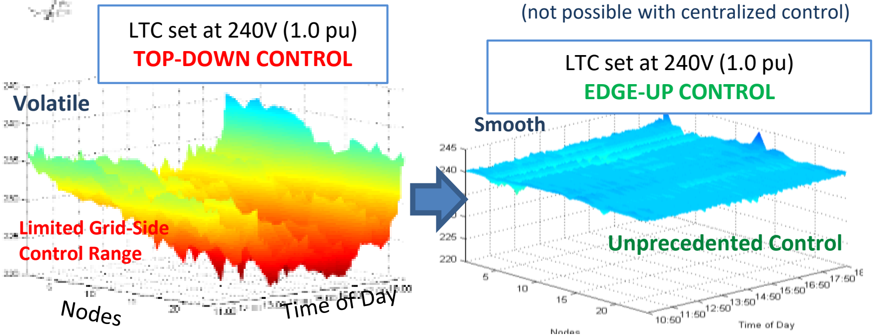
0-10 kVAR injection



Proven at 20+ utilities

- 5 MW 12 mile line
- 421 **Transformers**
- 4760 KVA
- 91 * 10 kVAR **Units**

Grid-Edge VAR Control
(not possible with centralized control)



Source: Southern Company and Varentec

Hybrid Transformers: *P/Q/V/I/Z control*

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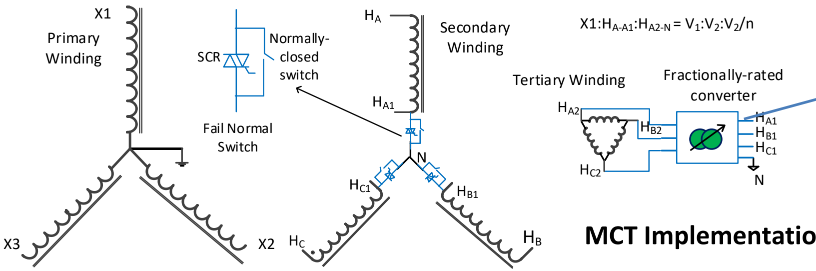
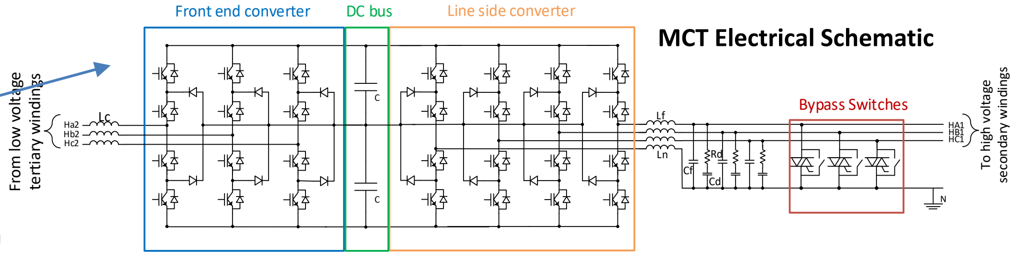
Standard Transformer Augmented With Fractionally-Rated (8%) Converter

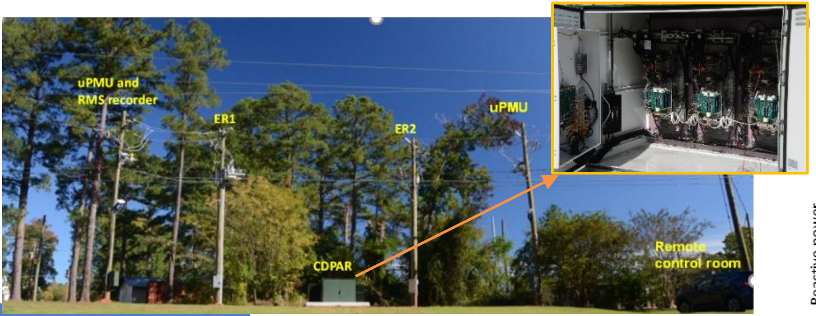
DOE Funded



MCT Implementation

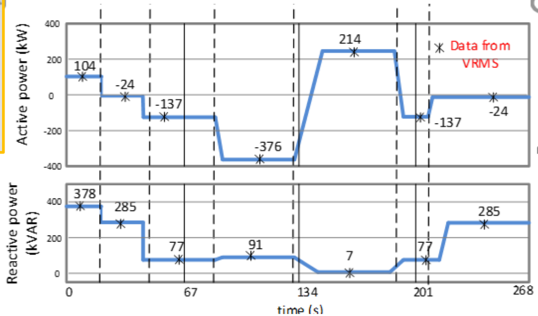


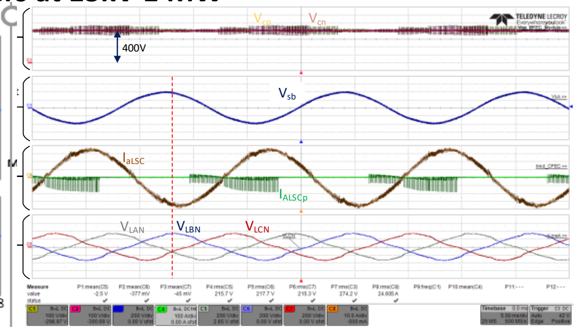
MCT Electrical Schematic



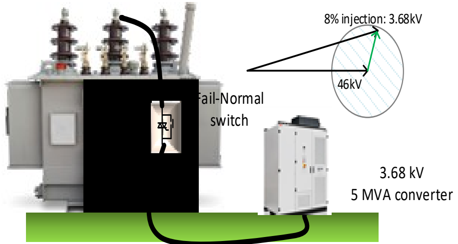
13 kV 1 MW field demonstration of power router

P control between 2 feeders at 13kV 1 MW





Standard Transformer 115/46 kV 60 MVA Transformer



Modular Controllable Transformer – 60 MVA

- ✓ Uses standard large power transformer and widely available BTB converter
- ✓ Allows scaling to 100's of kV and 100's of MW with low-cost & small footprint
- ✓ Fail-normal approach retains basic functionality in case of converter failure.
- ✓ High impact at 10,000 bus system and Texas system (simulations)
- ✓ **Reduces cost and size for P/Q/V/I/Z control by 6X as compared with HVDC Light**

Solid State Transformers – Holy Grail!

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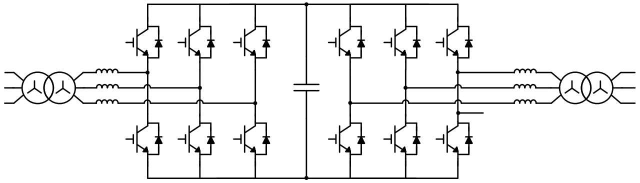
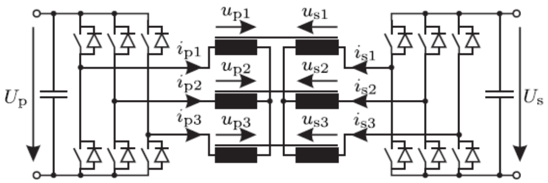
There is a need for bidirectional flexible scalable multi-port converters

Desirable SST Features:

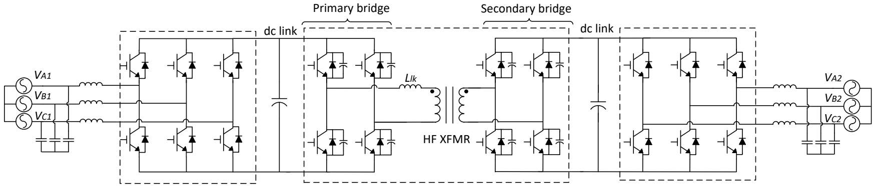
- Bidirectional & multi-port
- Sinusoidal waveforms
- Compact
- High bandwidth control
- Low EMI
- Transformer leakage management
- Modular and scalable
- BIL (grid connected)
- Fault current sourcing
- Robust



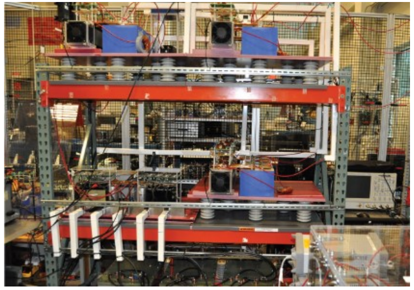
BTB Converter w/ 60 Hz Transformer



DAB Converter 5 kV 5 MW DC/DC

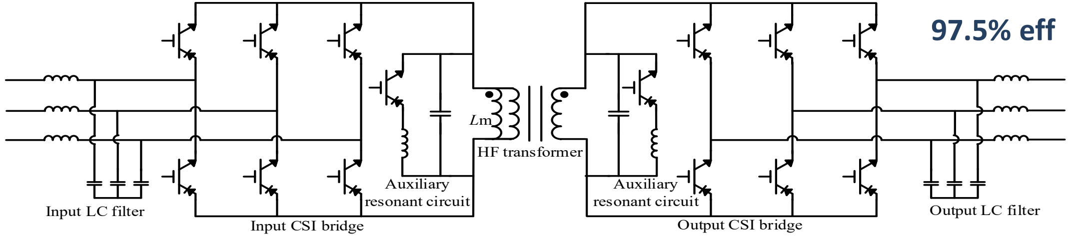


SST w/ DAB HF Link – 13 kV



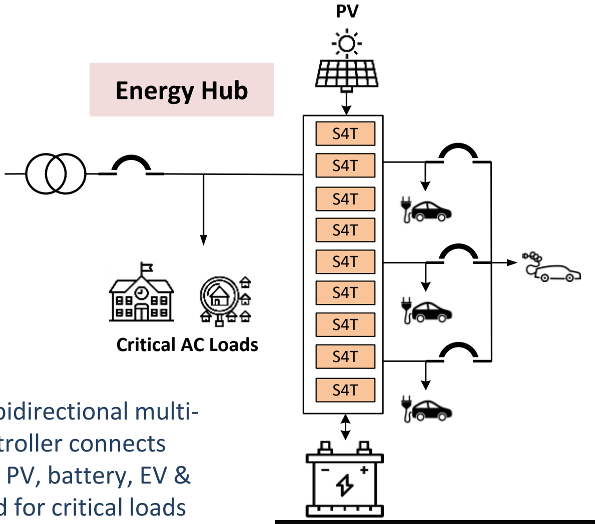
Soft Switching Solid State Transformers (S4T)

Bidirectional multiport converter with AC or DC input/output

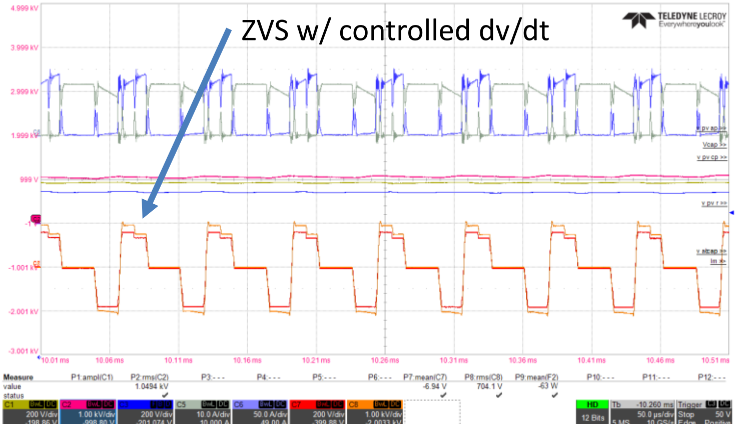


- **25 kVA building block, 97% eff.**
- **480-600 VAC, 600-800 VDC**
- **Triport – DC+DC+AC or AC+AC+DC**
- **Parallel to multi MW level**

- **HF transformer isolation**
- **Bidirectional universal converter**
- **Sinusoidal/filtered input/output**
- **ZVS, low dv/dt, low EMI, CSI**

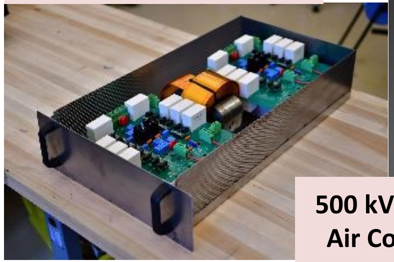


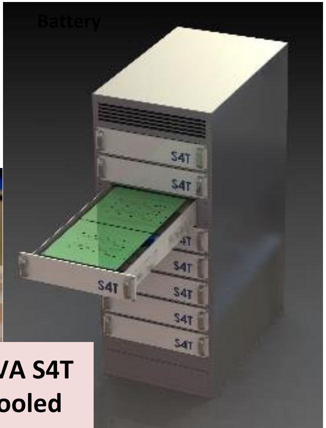
Flexible bidirectional multi-port controller connects grid with PV, battery, EV & microgrid for critical loads



S4T Applications:
- 5 kV MVDC networks
- 7.2 kV 50 kVA SST
- 300 kW MVSI PV Farm
- DC Fast Charging
- PV/Storage/Grid
- Data Center/UPS

25 kVA-v1 50 kVA – v2 S4T Module Air Cooled





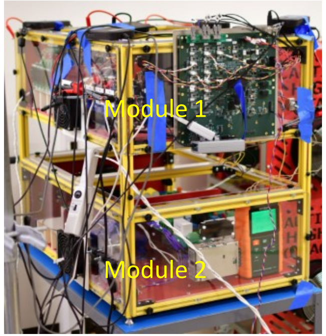
500 kVA S4T Air Cooled

S4T - MV Applications

Microgrid Power Conditioning Modules

- 50 kW MVDC (5 kV DC) to LVAC/LVDC (480 V AC / 600 V DC)
- Applications: Microgrid, industrial plants, DC distribution
- 3.3 kV SiC devices and 1.2 kV devices (Si or SiC)
- DC scaling – control in low-inertia & series stacking
- Characterizing 3.3 kV SiC reverse blocking module
- Funded by **Power America**

3300 V/45A
5 RB-SiC devices

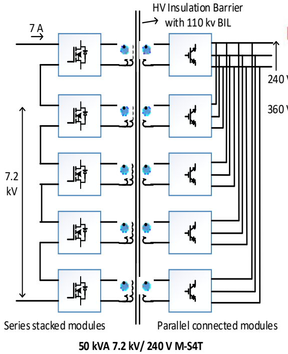


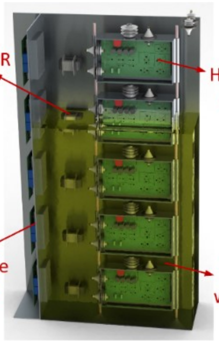
25 kW 16 kHz
4:1, 55 kV BIL

50 kVA MVDC/ LVDC or LVAC

Modular Solid State Transformer

- 50 kVA 7.2 kV 1Ph AC to 240VAC &360VDC
- 3.3 kV SiC devices and 1.2 kV devices (Si or SiC)
- Applications: Retrofit SST, Grid Energy storage integration, Data Center distribution, Shipboard Applications, traction etc.
- 1Ph AC scaling - series stacking
- >90 kV BIL management
- Funded by **ARPA-E CIRCUITS**

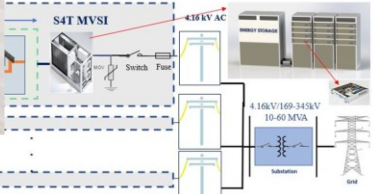




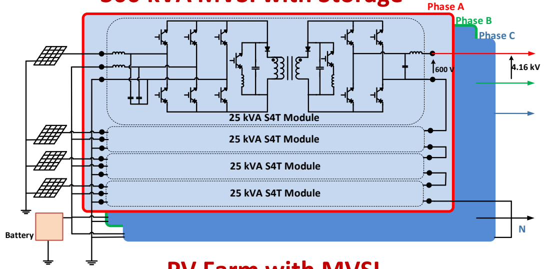
Oil cooled M-S4T with BIL management

MVSI for Solar PV farms

- 300 kVA MVSI to connect 1000V PV strings and energy storage to 4.16 kVAC distribution
- Applications: Utility scale PV farms with Storage
- 1.7 kV devices (Si or SiC)
- 1 Ph scaling – low-inertia & series stacking
- Control of distributed string inverters and storage
- Reduces LCOE and improves efficiency by 2.4%
- Funded by **DoE SunShot**



300 kVA MVSI with Storage



PV Farm with MVSI

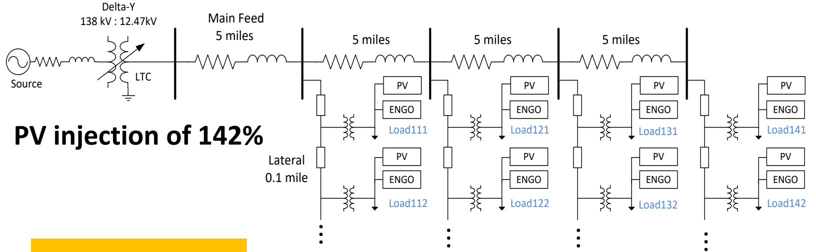
All Seems Good – So Where are the Challenges?

Control Issues for PE Grids

- Millions of DER inverters are being deployed while serious questions remain on modeling, control and grid integration
- PE converters designed as single-input single-output systems: multi-converter systems operate as master/slave
- Constant P control (MPPT) presents negative impedance - can destabilize systems under high DER penetration
- Droop based volt-VAR control uses the entire voltage band, and is problematic due to interactions between inverters
- Interacting inverter control loops, PLLs and controller design based on knowledge of system parameters - not scalable
- Need to look at a new paradigm under high DER penetration with thousands of grid-connected inverters

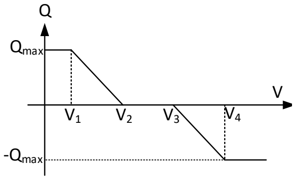
Concept of Collaborative Control is proposed for managing a large number of grid connected inverters

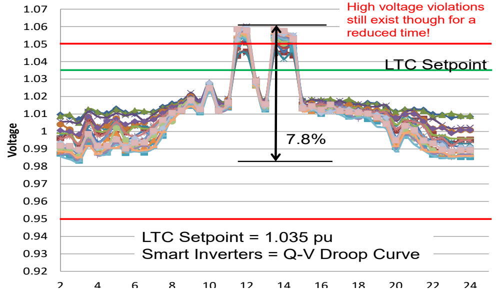
Typical distribution feeder with high PV penetration



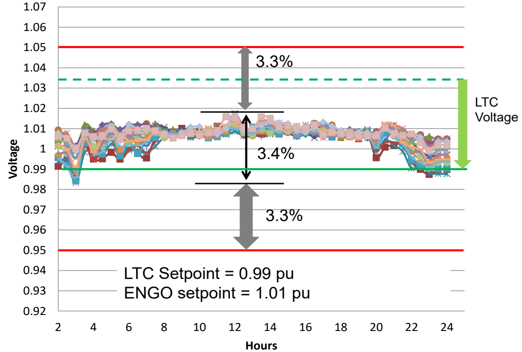
PV injection of 142%

Smart Inverters w/ Q-V Droop





Collaborative VVC – feeder simulation



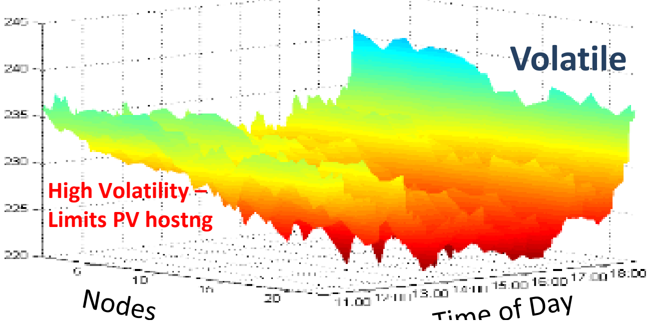
Collaborative Control

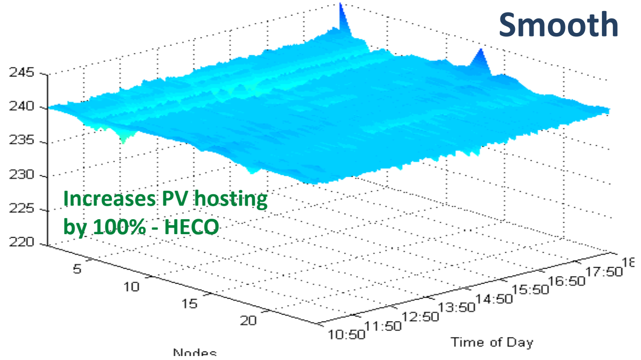
- ‘Collaborative Control’ is proposed for systems with many dynamically controllable edge devices (such as smart inverters)
- Edge-devices follow simple ‘rules’ (e.g. Vref) and act in real-time to fulfill individual objectives (to the extent possible)
- Accurate and real-time knowledge of system topology or state is not required – set point & slow comms for market function
- Individual device operation is based on local measurements, not on centralized state estimation, dispatch or communication
- Signaling between collaborating devices depends on deviations at point of coupling, a result of device and system interactions
- Collaborators have individual objectives, but can also provide ancillary support for system voltage/frequency & VARs
- In a multi-owner resource-constrained system, there is no guarantee that all objectives will be met at a point in time

Collaborative VAR control works – how about active power control ?

Distribution Line:	5 MW 12 mile
Service Transformers:	421
Total Loads:	4760 KVA
Decentralized VAR Units:	91 * 10 kVARs

Source: Southern Company and Varentec





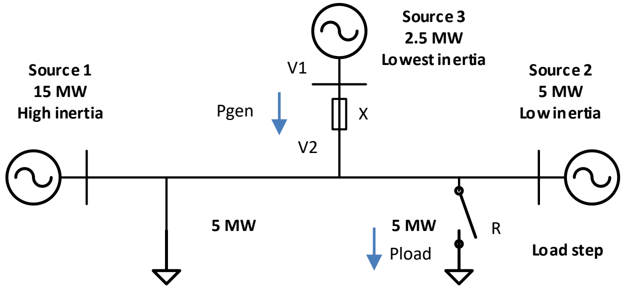
On-grid demonstration of collaborative VAR control

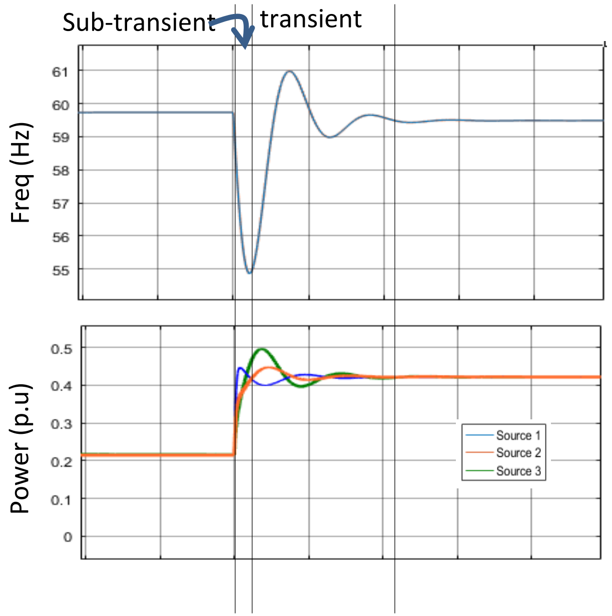
Fundamentals Revisited - Generators

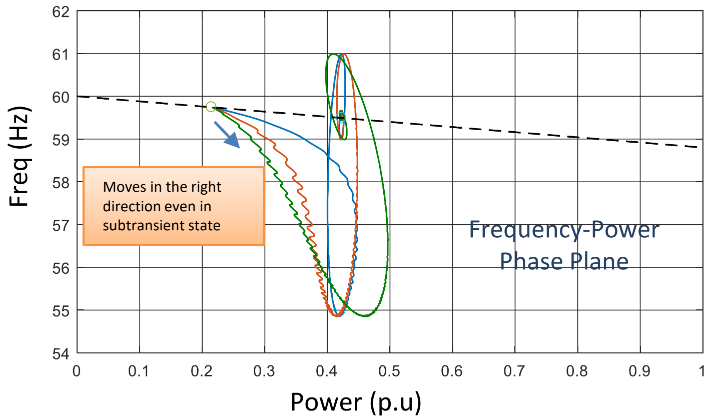
- Grid operation is based on generators as voltage sources, loads as impedances/current sources, and line impedances between sources

  $P_{gen} = (V1-V2) \sin \delta / X \qquad P_{load} = V^*V/R$

- Vline is approximately constant and follows P-F droop curve (system rule)
- Three stages of response
  - Subtransient – system (impedances, inertia) dependent – <u>signaling phase</u>
  - Transient – Governor control
  - Transactive – Negotiated end state (market)
- Generators have intrinsic sub transient response that is aligned with power sharing response (P-F droop curve)
- <u>Generator frequency is self-aligning during sub-transient and transient</u>

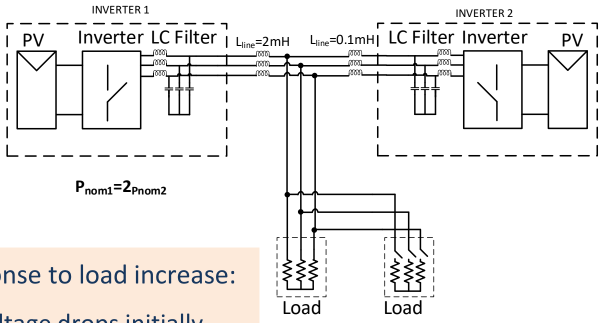






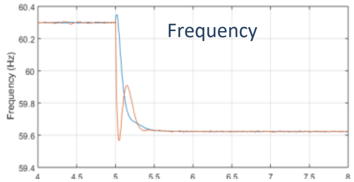
Fundamentals Revisited – Paralleled Inverters

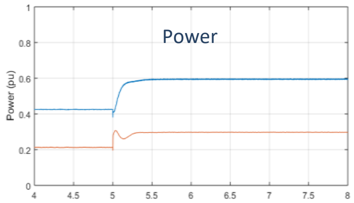
- Inverters operate with PLL and inner current loop, acting to control their individual outputs – works fine in grid following mode
- Multiple inverters acting simultaneously to ‘form’ the grid can lead to interactions & stability issues (unless in Master/Slave mode)
- In grid-forming mode, PV variability can create conditions of generation surplus or scarcity. How do we balance the system?
- Do we need to know network topology and generation/load states to manage the system – very challenging in multi-owner scenarios
- Do we need to know if there are any ‘grid-forming’ synchronous generators on the system – what happens if the inertia changes?
- Is the grid there or not? Do we need to be in grid-following or grid-forming mode? How quickly do we need to change modes?
- When frequency is changing quickly, how do we measure frequency? Are ‘phase’ or VAR even meaningful?
- With millions of inverters from many manufacturers over decades, lagging standards, and an unknown and changing network – we need to ‘guarantee’ stability (what does that really mean?).

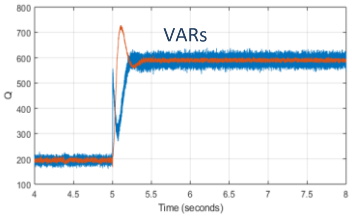


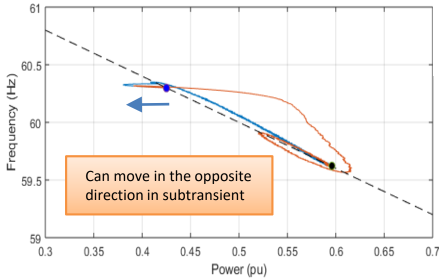
Response to load increase:

- ➢ Voltage drops initially
- ➢ PLL frequencies not aligned during transient → dynamic ‘VAR’ flows
- ➢ As all inverters try to set frequency, causes interactions & degrades system stability





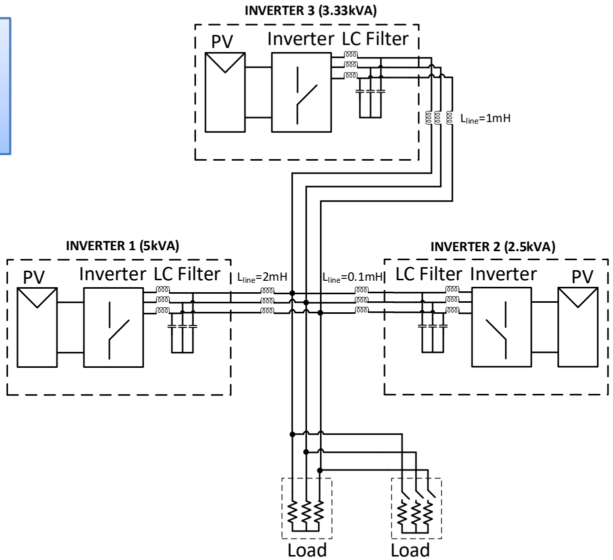


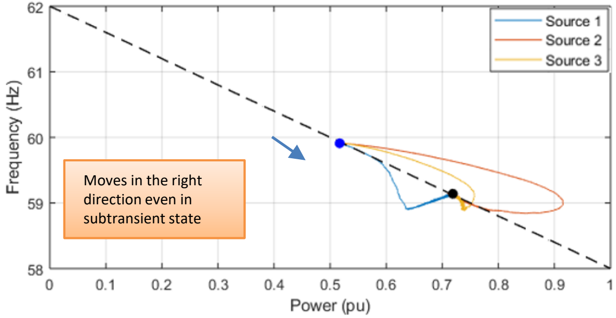


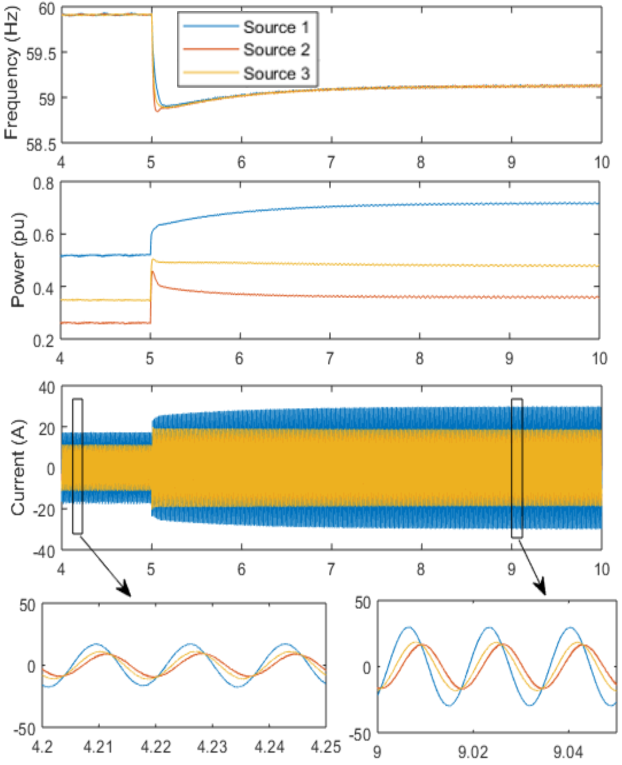
Intrinsically Grid Aligning Inverters
A New Concept based on Collaborative Control

3 inverters are controlled so they are dynamically self-aligning with the ‘local’ grid – transition to new P-F point w/ step load

- Universal ‘collaborative’ control - automatically operates in both ‘grid-following’ and ‘grid-forming’ modes
- Does not need - knowledge of topology, coordination between inverters, or whether system is grid connected or not
- Inverter transactive control is based on P-F droop, and can work in hybrid systems (w/ synch gen)
- System can do black-start, voltage control, VAR support – no PLL needed







System Issues

Power Electronics on the Grid – More Questions

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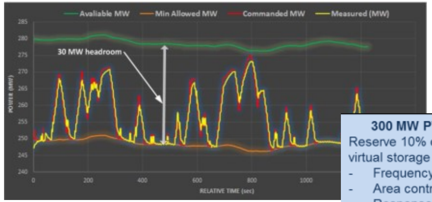
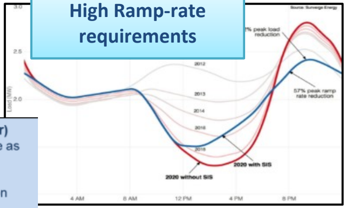
System Stability

- Can we ensure stable system operation under high penetration of DER?
- Do we need more inertia? How about hybrid (generator + PE) systems?
- Should inertia be in the form of energy storage? Or virtual inertia?
- Can we model system stability with current simulation tools?



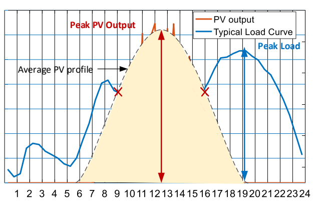
300 MW PV plant (First Solar)
Reserve 10% of headroom for use as virtual storage (why MPPT?)
- Frequency regulation
- Area control error minimization
- Response to AGC signal

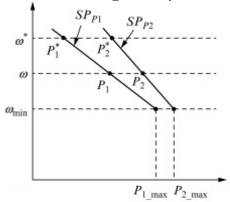
High Ramp-rate requirements



Dynamic Balancing

- How do you control a system as it moves between supply surplus/scarcity?
- Can active and reactive power be balanced in a decentralized fashion?
- Can we know available capacity before connecting a load to the grid ?
- Can dynamic balancing be achieved with millions of intelligent prosumers?





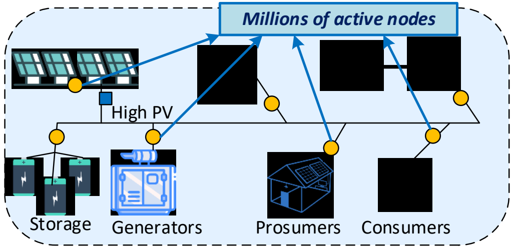
Varying resources needs a 'load-follows generation' approach

Droop is used to negotiate power sharing – decentralized!

Multi-Owner Architecture

- Can we have a real-time market for millions of prosumers?
- Can we meet personal goals while globally stabilizing the system?
- How can gaming be managed?
- Can this be decentralized and feature an open ledger?

- Millions of active nodes
- Individual goals
- Satisfy global constraints
- Impossible to centrally manage system



Grid of the future

Cyber-Security, Communications & Resiliency

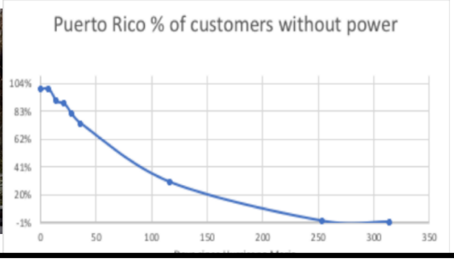
- Can smart inverters reduce risk of large outages due to cyber attacks?
- Can system integrity and stability be ensured with loss of comms?
- Can we restore power quickly after HILF events?
- Are bottom-up black start and clustered microgrids viable?

Reports of an unprecedented grid "cyber event" caused a stir last week in power sector and cybersecurity circles [1]

[1] "Experts assess damage after first cyberattack on U.S. grid", E&E NEWS, available at: https://www.eenews.net/stories/1060281821

Resilient Systems

Today's Grid - Rigid

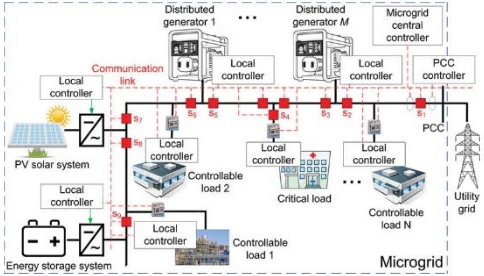


Resilient Grids (or not!)

Post Hurricane Maria – 250 days in PR; CA wild fire impact

Top-down restoration challenging for High Impact Low Frequency events

How can susceptible communities be designed with higher resiliency?



Advanced Grids

Manage increased penetration of DER & Microgrids

This is Distributed – not Decentralized

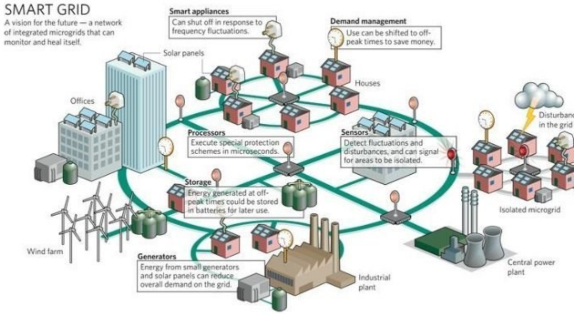
Future Decentralized Grid – Ad Hoc, Fractal & Flexible

Ad Hoc Grids

- Build flexible and resilient grids very quickly
- Incrementally add capacity as needed or available
- Standard building blocks that allow scaling
- Rapid deployment, mobile and transportable
- Resilient – survives and operates through major faults
- Simple – minimal technical competence in field

Agile Fractal Grids

- Distribution system – network of interconnected microgrids
- Preserves advantages of centralized power system
- Islands into microgrid clusters under severe faults - resilient
- Fractal control law allows operation as system fragments
- Resiliency achieved with reduced dependence on comms
- Desirable, but challenging with existing technology

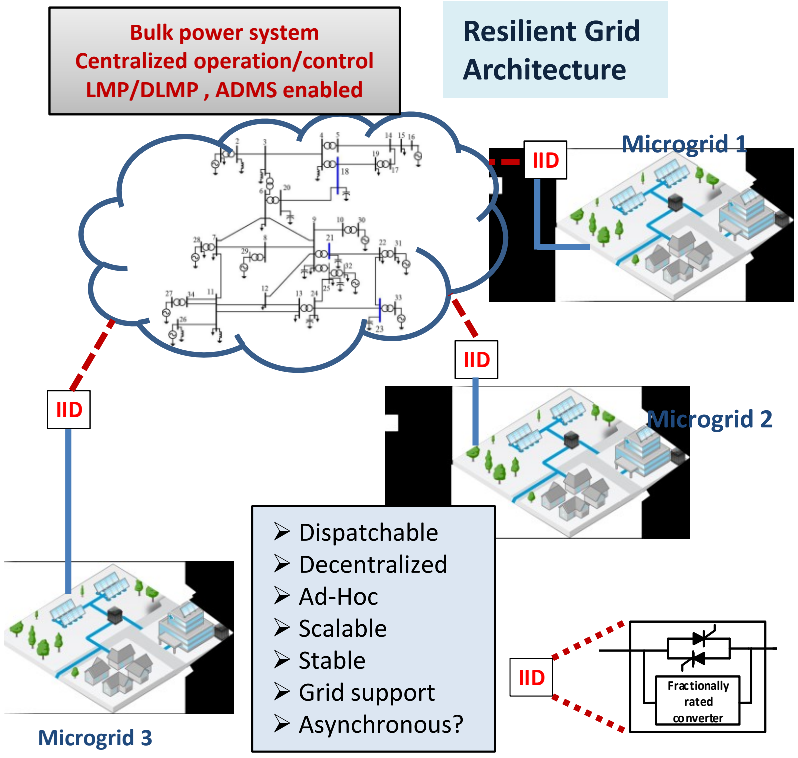


tloving | Jul 19, 2016 | In the News, The Agile Fractal Grid

Do we know how to implement decentralized grids – not very well. We don't even know all the questions.

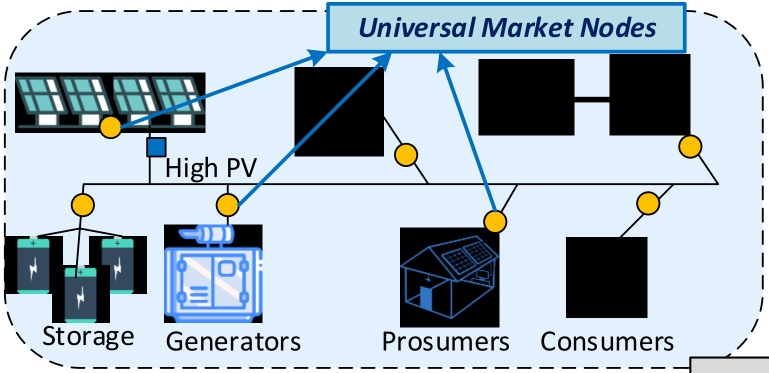
Decentralized Grid – Puzzling Questions

- Centralized bulk power system with islandable 'microgrids' improves system resiliency – connect/disconnect at will or on utility command
- How do individual devices know if they need to be in grid-following or grid-forming mode, or have to black-start (or not)? Is a PLL needed?
- Can we form a cluster of microgrids using 'bottom-up' black-start, and reconnect to the grid when desired – with minimal comms/control
- In islanded mode, if 90% of load is supplied by PV and 50% of the supply is lost – how does the system dynamically balance itself?
- How does a load know when there is sufficient capacity to connect to the grid, or if it needs to disconnect because supply is constrained?
- The volatility of PV/storage suggests the system will be in supply-constrained or supply-surplus modes – how do we know/coordinate?
- Each prosumer has different cost points, financial objectives and operating constraints – how do we coordinate and prevent gaming?
- No surprise that traditional microgrids are centralized with Real-Time coordination, adds to the cost – can this even be made decentralized?

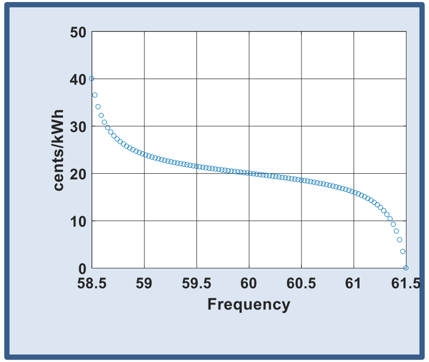


How do we implement such a decentralized flexible control?

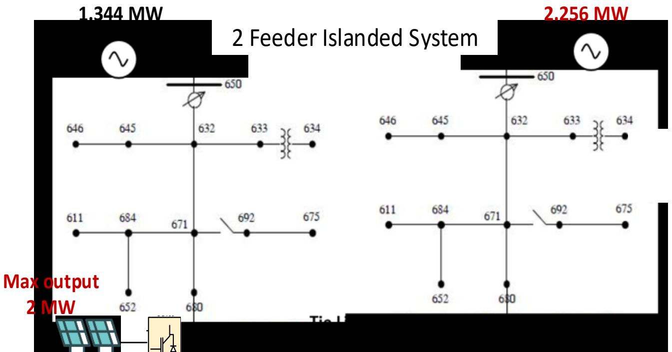
Decentralized Control of Microgrids/Grids



Off-grid community

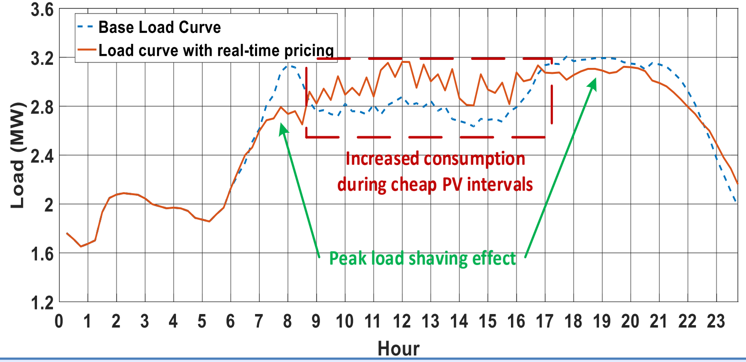


Global mapping for frequency vs real-time price – only control parameter



Distribution Feeder showing dynamic self-pricing operation

- Price-frequency droop curve allows integrated transactive-physical control
- Achieves dynamic balancing without communications or topology knowledge
- True multi-agent framework with nodes controlled on frequency -> no coordination
- System operates with heavy PV penetration to reflect volatility in PV rich grids



Loads/Sources Respond to Price

Novel concept for decentralized integrated physical and transactive grid control

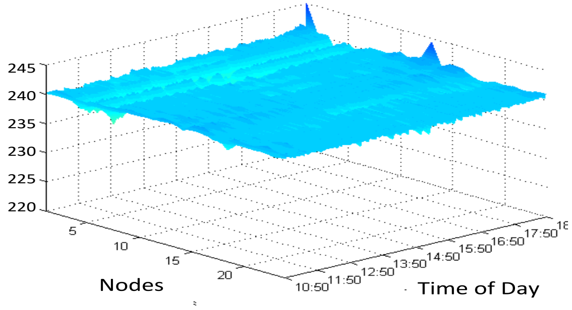
Simple Rules Manage Complex Systems

- The future grid needs to operate like a living ecosystem.
- Each node has intelligence and local visibility and influences the local environment based on simple rules for all nodes
- All nodes operate in their own self interest, but also act to sustain the system as a condition of market participation
- System is cyber-secure, self-aware & flexible, does not need information on network configuration, status & generation
- Real-time pricing information is derived at each node, allowing each node to optimize its behavior and investments
- Surplus/scarcity of resources triggers price swings that govern consumption, stabilize the system and drive investments
- Collaborative control and slow coordination allows the group/system to solve problems that an isolated node cannot
- Such a system is fractal, can be built from the bottom-up, and can realize high resiliency and availability at low cost

<u>Future Grid Attributes:</u>

- **Expandable**
- **Affordable**
- **Flexible**
- **Dispersed**
- **Secure**
- **Autonomous**
- **Dynamic-pricing**
- **Decentralized**
- **Resilient**
- **Simple**
- **Market**

Collaboration using simple rules in an ant colony



Grid stabilization using collaborative control

Grid Transformation is Coming…*Are We Ready?*

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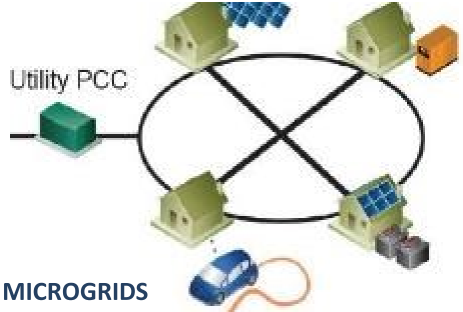
Current Grid	Future Grid
Centralized w/ Large Assets	Decentralized & Distributed
Planned & Scheduled	Ad-hoc & Variable
Coordinated, Dispatched	Autonomous, Self-Optimizing
Limited Observability & Poor Control	Smarts & Dynamic Control at Edge
Top-down, Structured, Fragile	Bottom-up, Fractal, Resilient
Grid as a Resource, Limited Market	Grid as an Ecosystem, Broad Market
Generation follows Load	Load follows Generation

This is a New Grid Paradigm – Power Electronics is Key



Enabling A Power Electronics Grid

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- The grid paradigm is rapidly changing, driven by exponential technologies that simultaneously impact many adjacent areas, making prediction very difficult
- The three main drivers are ‘digitalization’, ‘decentralization’ and ‘decarbonization’ – power electronics is integral to all three
- We are on a journey from a centralized system to a decentralized system with intelligence and dynamic control integrated into millions of grid-edge generation, storage & load devices
- An approach is needed that provides a glide-path from today’s system to the new system over the next decade
- Significant gaps remain in our understanding of such decentralized systems:
    - modeling, analysis & simulation of decentralized grids with smart grid-edge control
    - techniques to control of millions of power converters that work collaboratively
    - grid-connected converter design taking into account system protection/interactions
    - control of grid operation with widely varying generation/demand ratios
    - design of resilient systems with graceful degradation, including bottom-up microgrids
    - coordinated operation of all grid participants – grid as a living ecosystem
    - ability to operate without communications (post cyber or HILF events)

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Questions?

ddivan@gatech.edu